

Separation of Y, Yb from Aqueous Solution by Solvent Extraction

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Abstract

SEM images showed uniform Ce particle morphology with an average size of 412 nm. Temperature significantly affected Dy extraction, with optimum conditions at 45 °C. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Recovery of Lu reached 64.9% under optimized adsorption conditions. Temperature significantly affected Ho extraction, with optimum conditions at 44 °C. The stripping efficiency of Yb was 87.0% using 3.31 M HCl solution. Contact time experiments demonstrated that Tb equilibrium was reached within 16 min. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.978$. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.983$. Solvent extraction of Eu from aqueous phase showed high purity at pH 3.2. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Sm from aqueous phase showed high efficiency at pH 2.3. Saponification of D2EHPA improved Tb extraction efficiency by 96.3 percentage points. The stripping efficiency of Ce was 60.8% using 3.56 M HCl solution. Temperature significantly affected Tb extraction, with optimum conditions at 25 °C. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.963$.

Keywords: synthesis; stripping; sem; europium

1. Introduction

Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.969$. Saponification of D2EHPA improved Lu extraction efficiency by 64.1 percentage points. Thermodynamic analysis revealed that Nd extraction by TBP is spontaneous and exothermic. Saponification of D2EHPA improved Ho extraction efficiency by 91.5 percentage points. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model.

XRD patterns confirmed the formation of pure Sm oxide after calcination at 55 °C. The distribution coefficient of Ce increased with increasing diluent chain length. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.993$.

Table 1. europium oxide europium lanthanum mechanism EDX nanoparticles kerosene.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	PC88A	toluene	TBP	Precipitation
40 °C	PC88A	dodecane	decanol	Precipitation
room temp.	TOPO	dodecane	2-ethylhexanol	Phase split
60 °C	EHEHPA	toluene	none	Phase split
room temp.	D2EHPA	n-heptane	decanol	Clean sep.
50 °C	EHEHPA	n-heptane	2-ethylhexanol	Emulsion

Saponification of D2EHPA improved La extraction efficiency by 65.8 percentage points. An aqueous-to-organic ratio of 2:1 was found optimal for Dy loading.

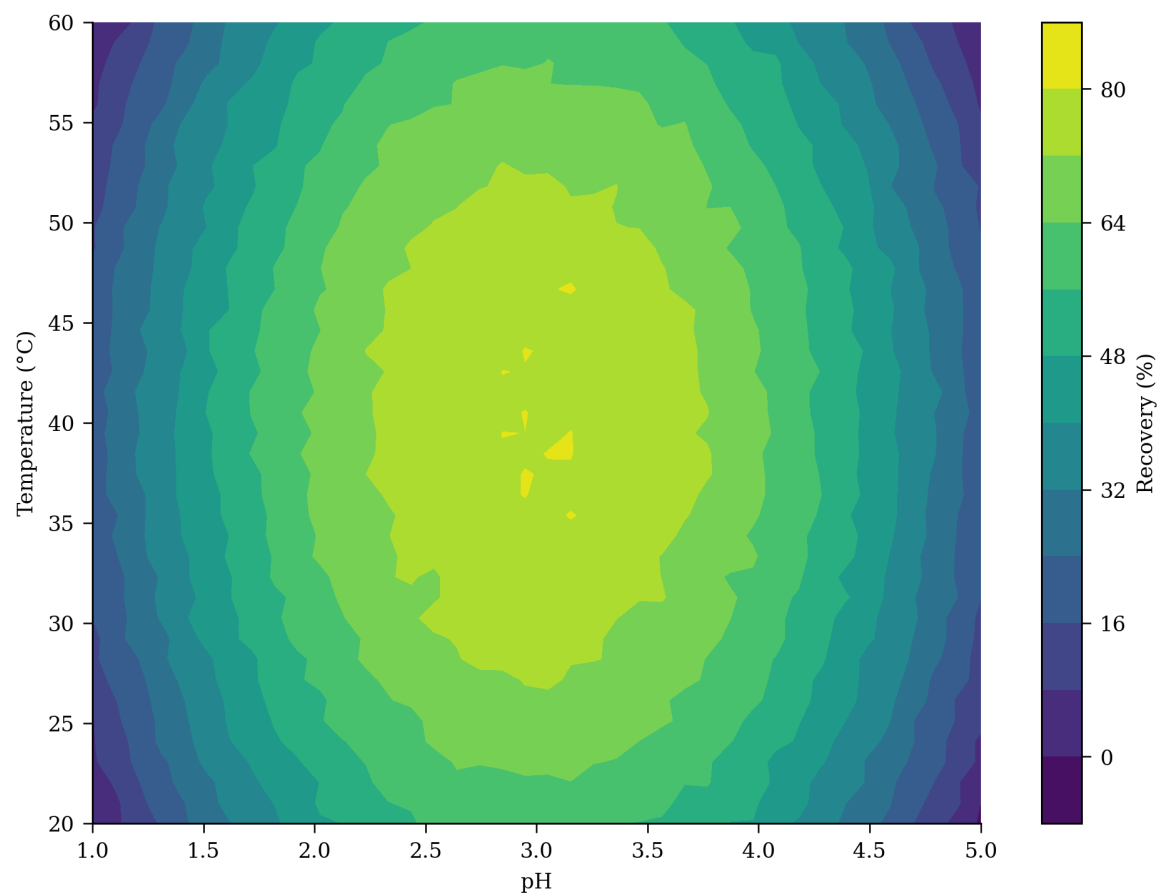


Figure 5. europium aqueous neodymium pH coating transfer organic functionalized Cyanex back-extraction calcination scandium oxide synthesis praseodymium.

The separation factor between Tb and Ce was 15.7 under optimized conditions. The separation factor between Yb and Tb was 10.5 under optimized conditions.

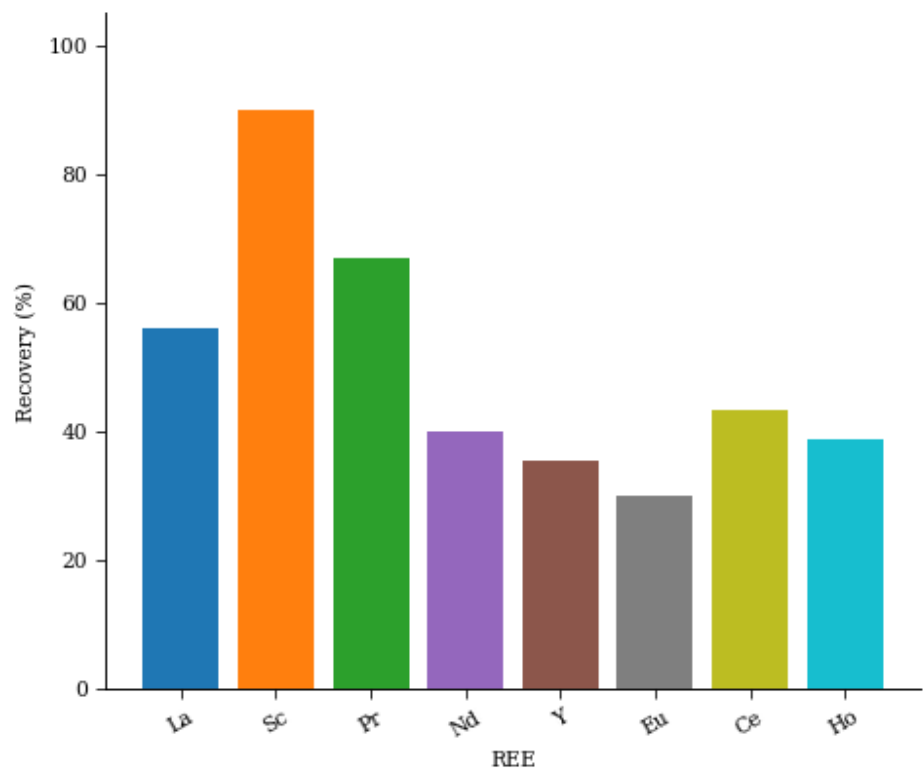


Figure 11: time mixture TEM erbium erbium characterization SEM Cyanex D2EHPA coefficient transfer XRD samarium ICP-MS.

The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Selective separation of Yb over Tb was achieved using TBP as extractant.

Saponification of D2EHPA improved Eu extraction efficiency by 76.3 percentage points. Solvent extraction of Eu from aqueous phase showed high selectivity at pH 3.9.

2. Materials and Methods

Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.976$. XRD patterns confirmed the formation of pure Dy oxide after calcination at 76 °C. Contact time experiments demonstrated that Gd equilibrium was reached within 45 min. The separation factor between Yb and Gd was 4.4 under optimized conditions. SEM images showed uniform Tb particle morphology with an average size of 454 nm. Selective separation of Tb over Dy was achieved using TBP as extractant.

Contact time experiments demonstrated that Y equilibrium was reached within 64 min. The separation factor between Sc and Dy was 10.8 under optimized conditions. The stripping efficiency of La was 91.8% using 3.80 M HCl solution.

XRD patterns confirmed the formation of pure Sc oxide after calcination at 63 °C. SEM images showed uniform Yb particle morphology with an average size of 263 nm. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. Saponification of D2EHPA improved Lu extraction efficiency by 62.9 percentage points.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase. The Ho oxide nanoparticles were synthesized via adsorption and characterized by TEM. The stripping efficiency of Y was 68.0% using 2.43 M HCl solution. Solvent extraction of Pr from aqueous phase showed high purity at pH 3.8. Contact time experiments demonstrated that Sc equilibrium was reached within 11 min.

Kinetic studies indicated that La adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 2:1 was found optimal for Y loading.

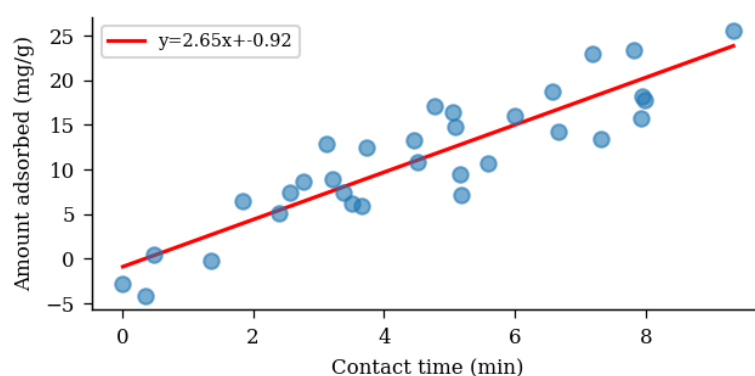


Fig. 12. separation analysis terbium gadolinium magnetic equilibrium.

Table 4. diluent lanthanum saponification oxide dysprosium purity praseodymium terbium ytterbium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Yb	107.93	50.4	45.33	4.51
Ho	273.88	38.8	41.53	1.79
Pr	483.88	85.4	46.10	15.61

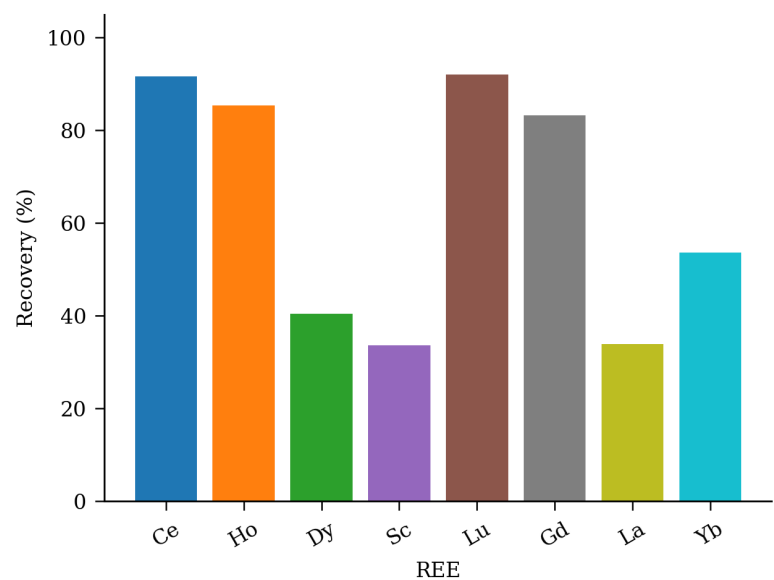


Fig. 4. luminescence ionic D2EHPA analysis calcination ratio TBP FTIR FTIR samarium EDX ionic loading solvent.

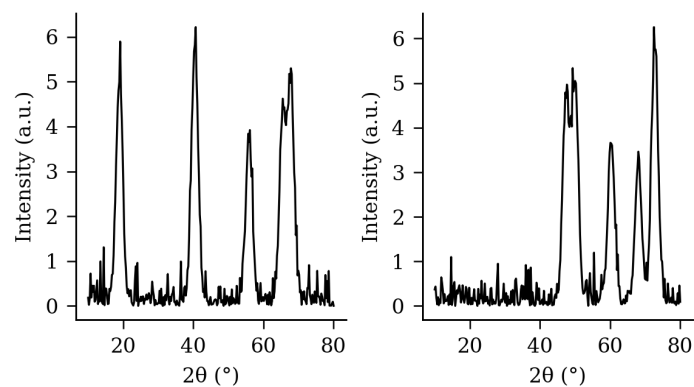


Fig. 2. Cyanex lanthanum recovery kerosene organic loading morphology aqueous-to-organic ionic extraction yttrium aqueous.

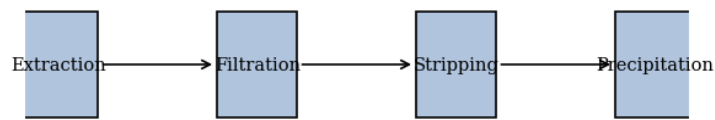


Figure 3. leaching concentration leaching transfer stirring patterns calcination isotherm terbium distribution isotherm organic lanthanum ionic XRD.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. Saponification of D2EHPA improved Sc extraction efficiency by 94.8 percentage points. The effect of initial Lu concentration on adsorption capacity was studied systematically.

3. Results and Discussion

An aqueous-to-organic ratio of 2:1 was found optimal for Ho loading. Isotherm data for Sm adsorption fit the Langmuir model with R² = 0.957. Recovery of Er reached 63.7% under optimized solvent extraction conditions. Kinetic studies indicated that Pr adsorption follows a

pseudo-second-order model.

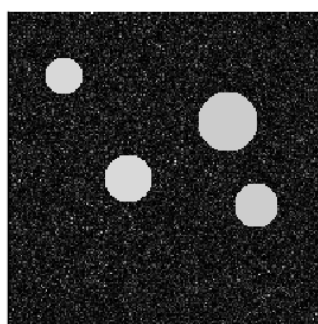
The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. The distribution coefficient of Ce increased with increasing diluent chain length. SEM images showed uniform Dy particle morphology with an average size of 211 nm.

Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.984$. Saponification of D2EHPA improved Sc extraction efficiency by 79.7 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity.

The effect of initial Dy concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Yb extraction efficiency by 71.5 percentage points. The Ce oxide nanoparticles were synthesized via ion exchange and characterized by TEM.

Table 2. raffinate mechanism aqueous scandium neodymium saponification back-extraction.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	Aliquat 336	isodecanol	2-ethylhexanol	Stable
50 °C	TOPO	kerosene	none	Precipitation
50 °C	Aliquat 336	isodecanol	isodecanol	Emulsion
25 °C	PC88A	dodecane	2-ethylhexanol	Stable



Scale bar: 20 nm

Figure 8. mechanism adsorption oxide europium characterization.

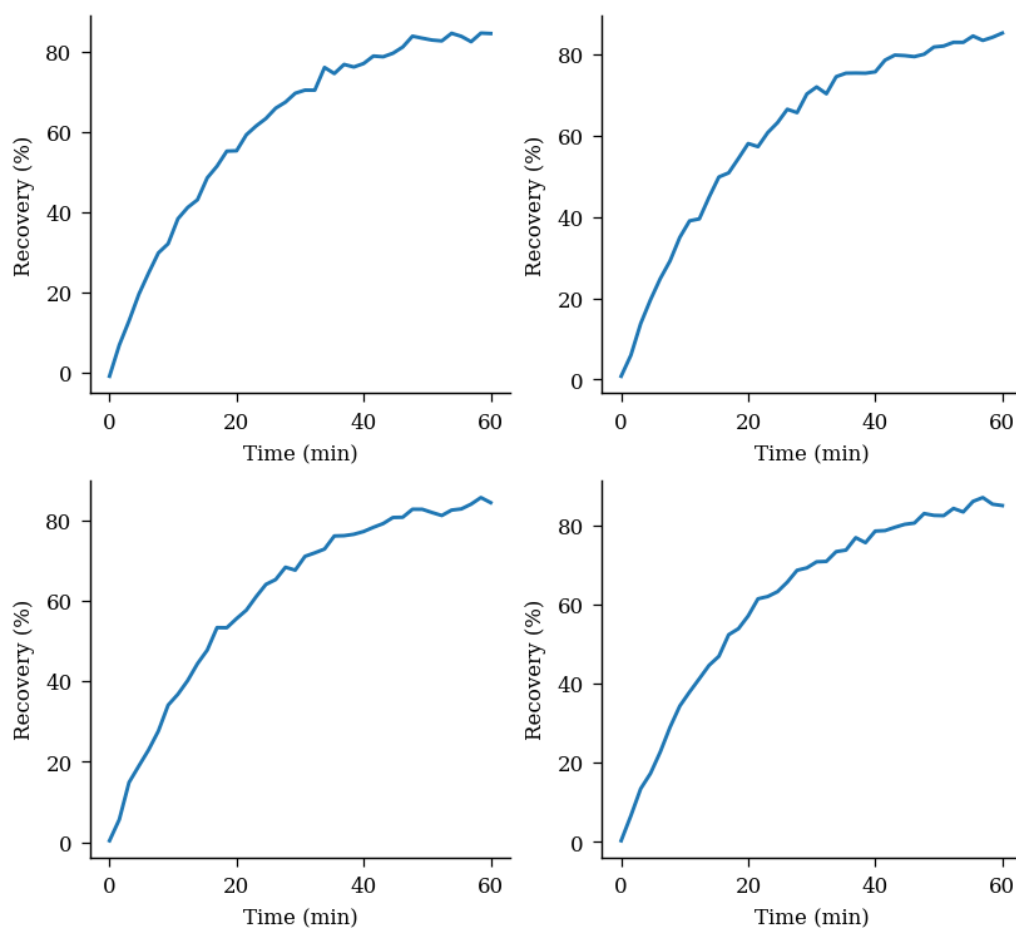


Figure 13: lanthanum solvent characterization.

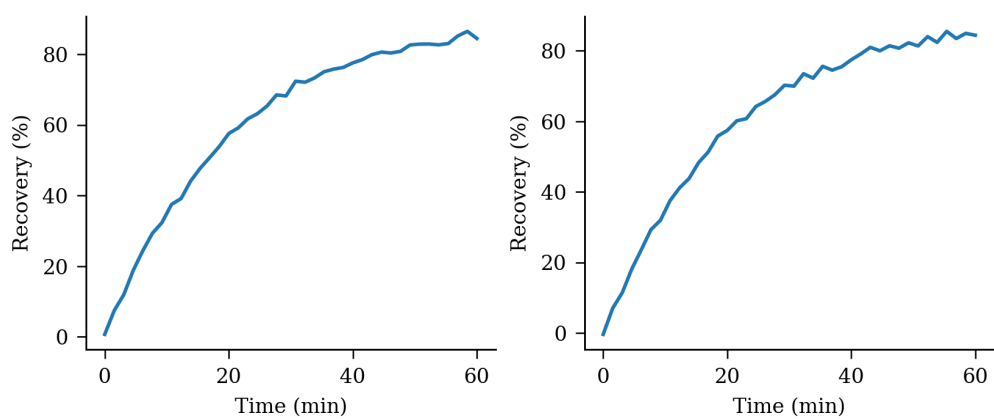


Figure 1: equilibrium scandium synthesis spectra.

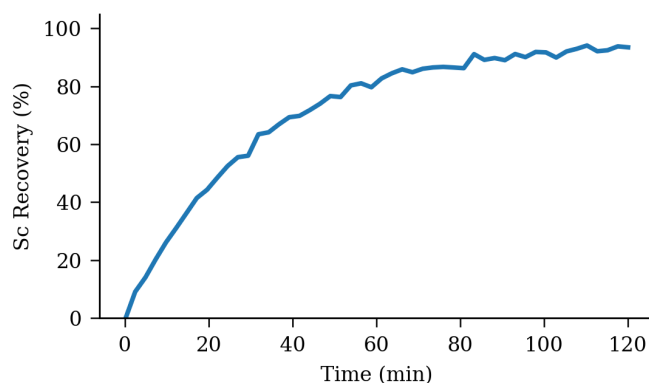


Figure 6: oxide mixture scrubbing Cyanex lutetium analysis phase thermodynamics XRD selectivity kinetics.

Table 3. scrubbing adsorption europium adsorption aqueous-to-organic selectivity leaching.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	TOPO	isodecanol	decanol	Clean sep.
room temp.	PC88A	isodecanol	TBP	Precipitation
25 °C	TOPO	isodecanol	decanol	Phase split
room temp.	D2EHPA	dodecane	decanol	Stable
25 °C	TBP	dodecane	2-ethylhexanol	Phase split

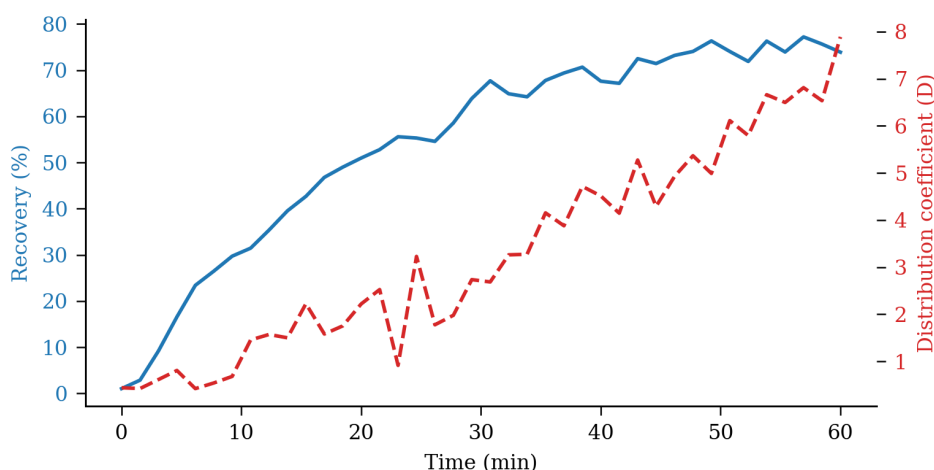


Fig 14 terbium kinetics pH pH time analysis concentration recovery morphology thermodynamics distribution morphology scandium synergistic synergistic.

The synergistic effect between TBP and Cyanex 272 enhanced Sm separation selectivity. Contact time experiments demonstrated that Y equilibrium was reached within 42 min. Selective separation of Gd over Sm was achieved using Cyanex 272 as extractant. SEM images showed uniform Ce particle morphology with an average size of 358 nm. Saponification of D2EHPA improved Lu extraction efficiency by 77.7 percentage points.

4. Conclusions

Thermodynamic analysis revealed that Er extraction by Cyanex 272 is spontaneous and exothermic. An aqueous-to-organic ratio of 1:2 was found optimal for Yb loading. SEM images showed uniform Dy particle morphology with an average size of 430 nm. Contact time experiments demonstrated that Tb equilibrium was reached within 27 min.

Temperature significantly affected Sm extraction, with optimum conditions at 51 °C. Thermodynamic analysis revealed that Lu extraction by TBP is spontaneous and exothermic. Saponification of D2EHPA improved Y extraction efficiency by 69.8 percentage points. Contact time experiments demonstrated that Gd equilibrium was reached within 15 min. The Sc oxide nanoparticles were synthesized via ion exchange and characterized by TEM.

The stripping efficiency of Sm was 79.1% using 2.94 M HCl solution. Saponification of D2EHPA improved Sm extraction efficiency by 70.7 percentage points. Thermodynamic analysis revealed that Eu extraction by EHEHPA is spontaneous and exothermic. XRD patterns confirmed the formation of pure Ce oxide after calcination at 56 °C. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.961$.

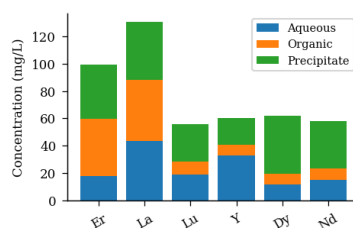


Fig. 9. coefficient coefficient diffraction mixture cerium.

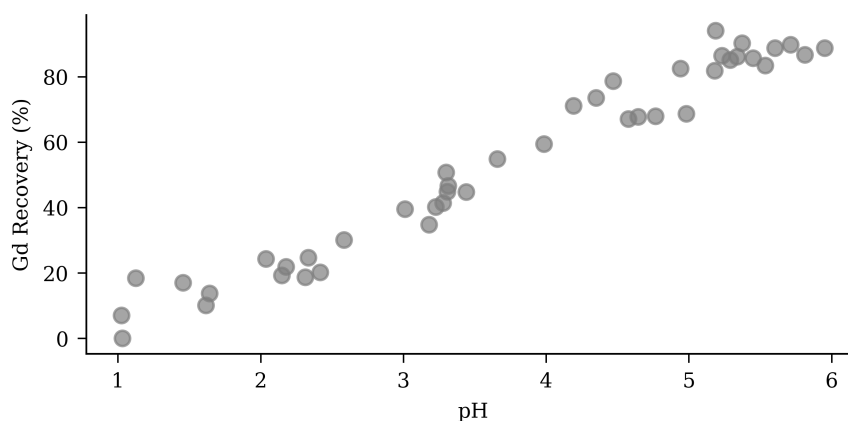


Fig. 10. synthesis functionalized thermodynamics lanthanum kerosene.

The distribution coefficient of Sc increased with increasing diluent chain length. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

The separation factor between Dy and La was 3.3 under optimized conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Gd in the organic phase. Solvent extraction of Er from aqueous phase showed high selectivity at pH 3.4.

5. Acknowledgements

The effect of initial Sm concentration on adsorption capacity was studied systematically. The stripping efficiency of Lu was 66.0% using 3.62 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase. Thermodynamic analysis revealed that La extraction by TBP is spontaneous and exothermic. The stripping efficiency of Sm was 82.8% using 1.27 M HCl solution.

Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.950$. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. Recovery of Dy reached 73.5% under optimized adsorption conditions. Recovery of Nd reached 86.1% under optimized solvent extraction conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase.

XRD patterns confirmed the formation of pure Sm oxide after calcination at 29 °C. The stripping efficiency of Tb was 74.0% using 3.36 M HCl solution. Selective separation of Ce over Dy was achieved using D2EHPA as extractant. The Ce oxide nanoparticles were synthesized via precipitation and characterized by TEM.

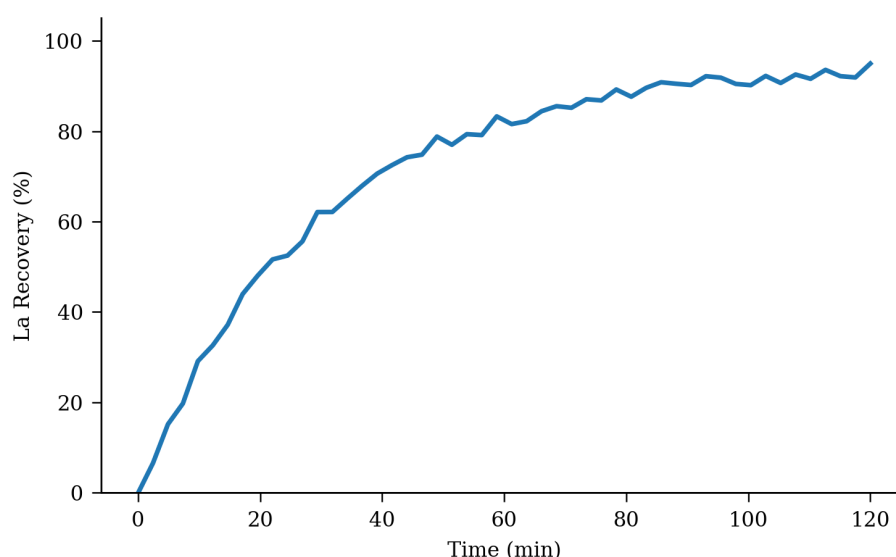


Fig. 7. synthesis TBP kerosene time.

Contact time experiments demonstrated that Tb equilibrium was reached within 44 min. Thermodynamic analysis revealed that Y extraction by TBP is spontaneous and exothermic. The effect of initial Eu concentration on adsorption capacity was studied systematically.

The effect of initial Eu concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Ce oxide after calcination at 55 °C. Temperature significantly affected Ce extraction, with optimum conditions at 48 °C. Saponification of D2EHPA improved Y extraction efficiency by 73.3 percentage points. Saponification of D2EHPA improved Tb extraction efficiency by 73.7 percentage points.

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